It's the elementary form of the mechanics of the world, and it does not need to mention "time." The world without a time variable is not a complicated one. It's a net of interconnected events, where the variables in play adhere to probabilistic rules that, incredibly, we know for a good part how to write. And it's a clear world, windswept and full of beauty as the crests of mountains; aridly beautiful as the cracked lips of the adolescent you loved.

ELEMENTARY QUANTUM EVENTS AND SPIN NETWORKS

The equations of loop quantum gravity[74] on which I work are a modern version of the theory of Wheeler and DeWitt. There is no time variable in these equations.

The variables of the theory describe the fields that form matter, photons, electrons, other components of atoms and the gravitational field—all on the same level. Loop theory is not a "unified theory of everything." It doesn't even begin to claim that it's the ultimate theory of science. It's a theory made up of coherent but distinct parts. It seeks to be "only" a *coherent* description of the world as we understand it so far.

The fields manifest themselves in granular form: elementary particles, photons, and quanta of gravity—or rather "quanta of space." These elementary grains do not exist immersed in space; rather, they themselves form that space. The spatiality of the world consists of the web of their interactions. They do not dwell in time: they interact incessantly with each other, and indeed exist only in terms of these incessant interactions. And this interaction *is* the happening of the world: it *is* the minimum elementary form of time that is neither directional nor linear. Nor does it have the curved and smooth geometry studied by Einstein. It is a reciprocal interaction in which quanta manifest themselves in the interaction, in relation to what they interact with.

The dynamic of these interactions is probabilistic. The probabilities that something will happen—given the occurrence of something else—can in principle be calculated with the equations of the theory.

We cannot draw a complete map, a complete geometry, of everything that happens in the world, because such happenings—including among them the passage of time—are always triggered only by an interaction with, and with respect to, a physical system involved in the interaction. The world is like a collection of interrelated points of view. To speak of the world "seen from outside" makes no sense, because there is no "outside" to the world.